

Quantifying Uncertainty in Marathon Finish Time Predictions

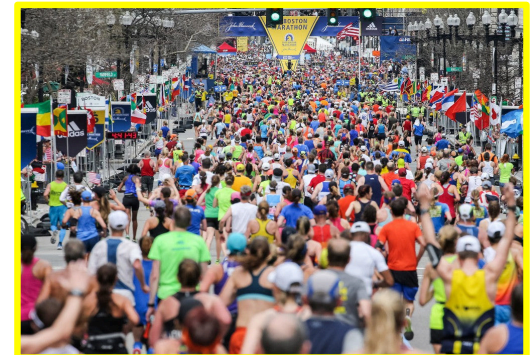
*Brandon Onyejekwe and Eric Gerber
Northeastern University*

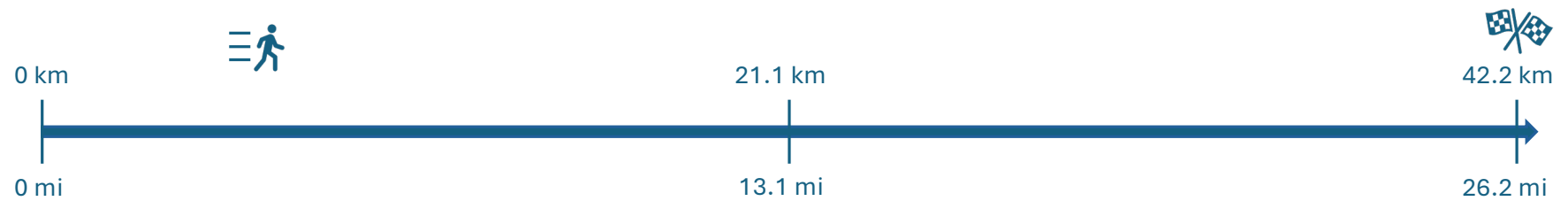




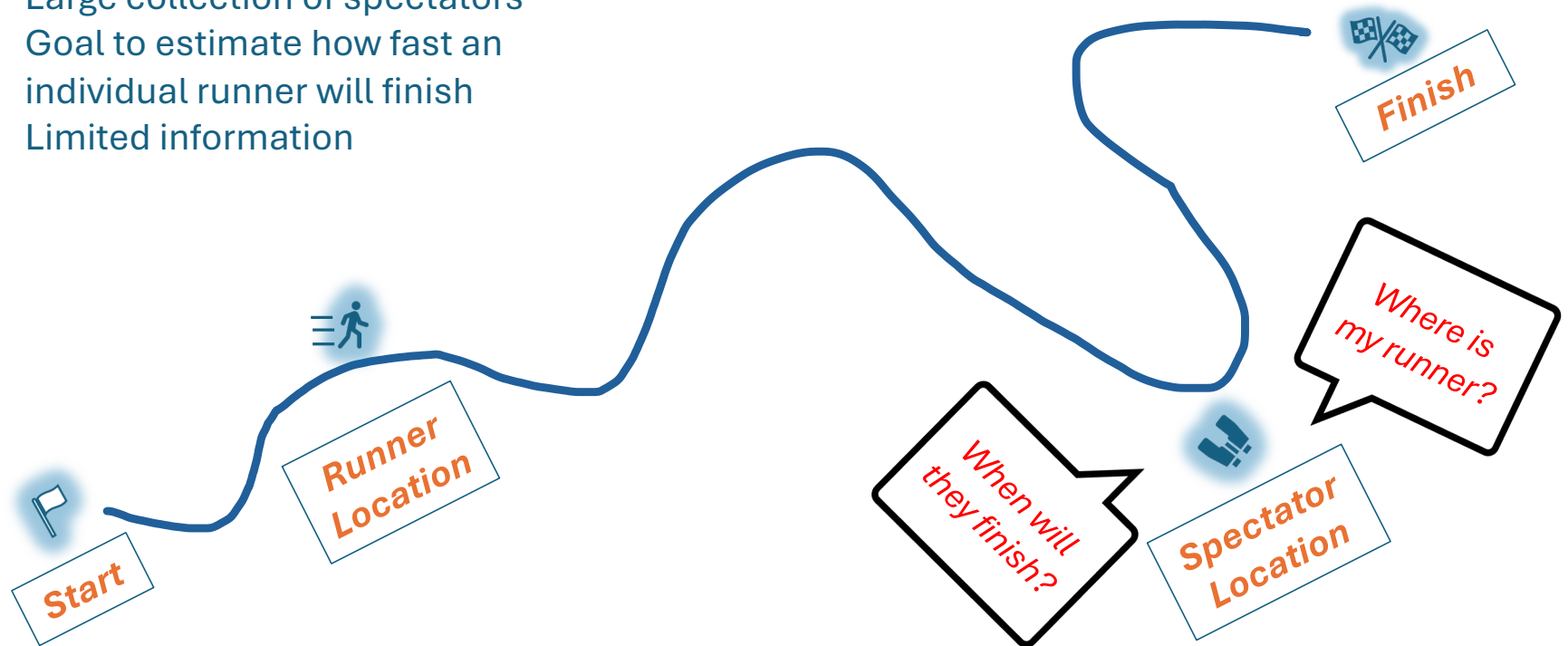
Marathons

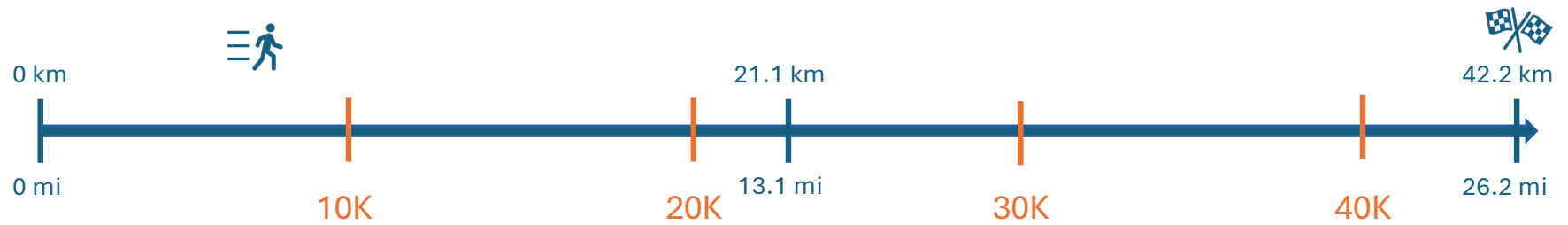
- long-distance road race
- 26.2 miles / 42.2 km
- Large collection of runners at different paces





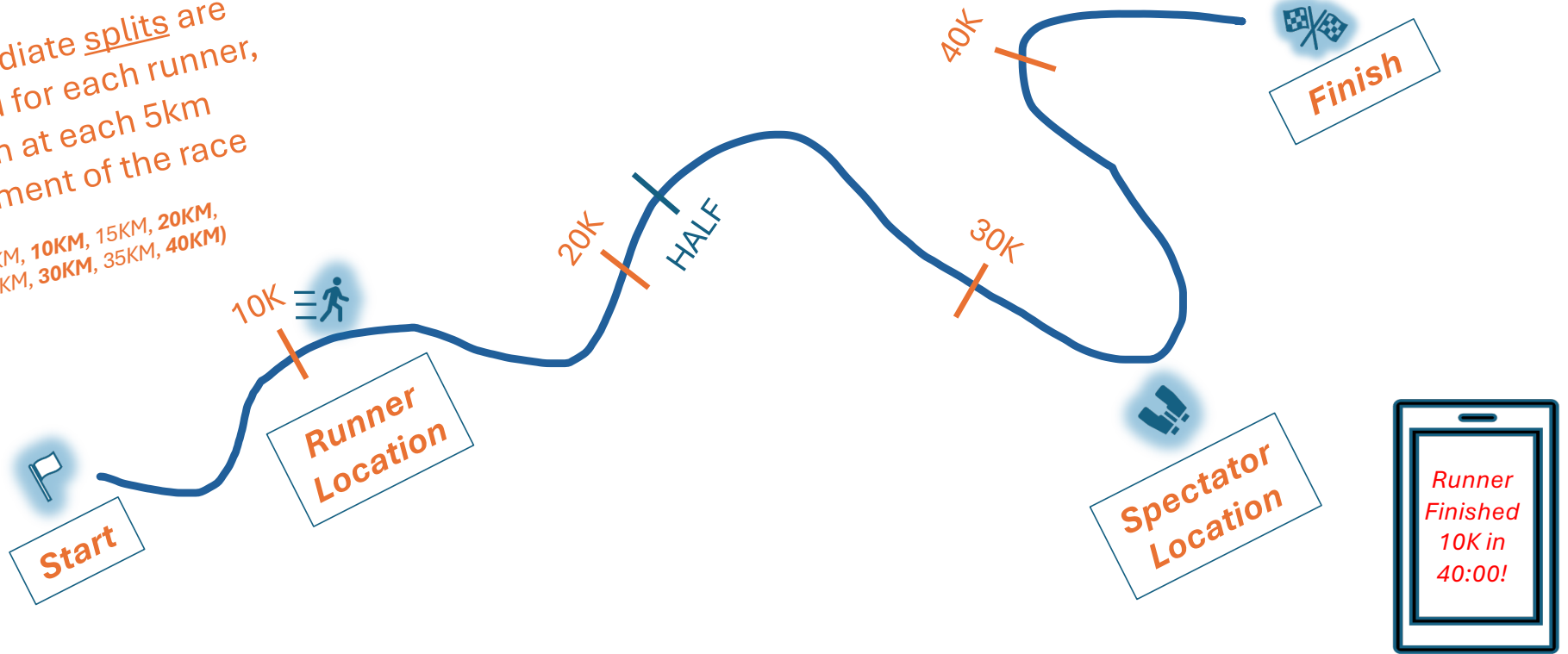
- Large collection of spectators
- Goal to estimate how fast an individual runner will finish
- Limited information

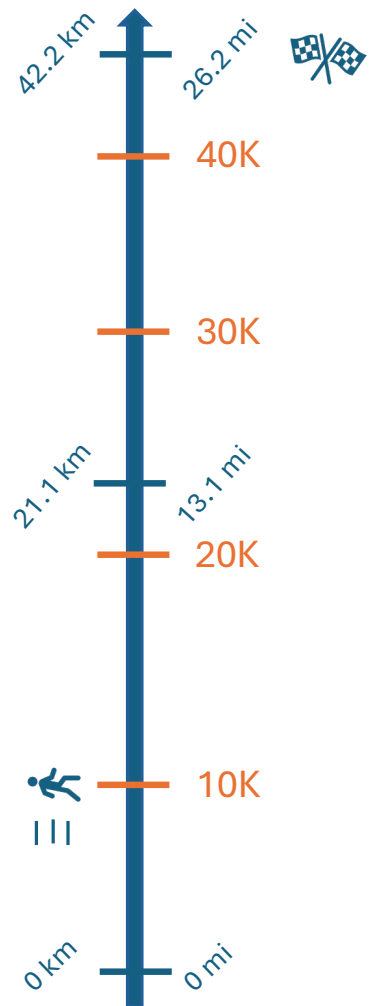




Intermediate splits are recorded for each runner, often at each 5km increment of the race

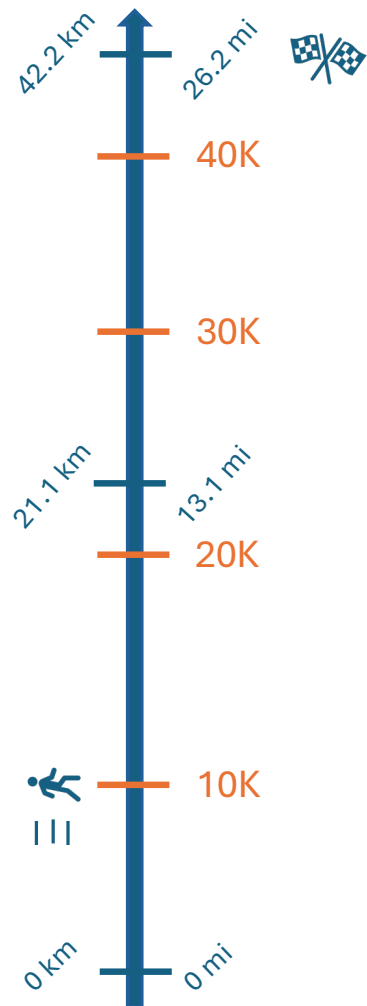
(5KM, 10KM, 15KM, 20KM, 25KM, 30KM, 35KM, 40KM)





Traditional methods to estimate finish times utilize the current **average pace** so far and extrapolate for the full distance, assuming that pace will be held constant for the rest of the race

<u>Distance</u>	<u>Time</u> (HH:MM:SS)	<u>Calculation</u>	<u>Prediction</u> At that stage*
After 10K	00:40:00	$\times (42.2/10.0)$	02:48:48
After 20K	01:21:45	$\times (42.2/20.0)$	02:52:30
After 30K	02:06:45	$\times (42.2/30.0)$	02:58:18
After 42K (finish)	03:01:20	$\times (42.2/42.2)$	03:01:20



Traditional methods to estimate finish times utilize the current **average pace** so far and extrapolate for the full distance, assuming that pace will be held constant for the rest of the race

What's Wrong With Traditional Predictions?

Problem 1: the estimates don't consider the **in-race context** that can determine if a runner is **likely to finish faster/slower than the observed pace**

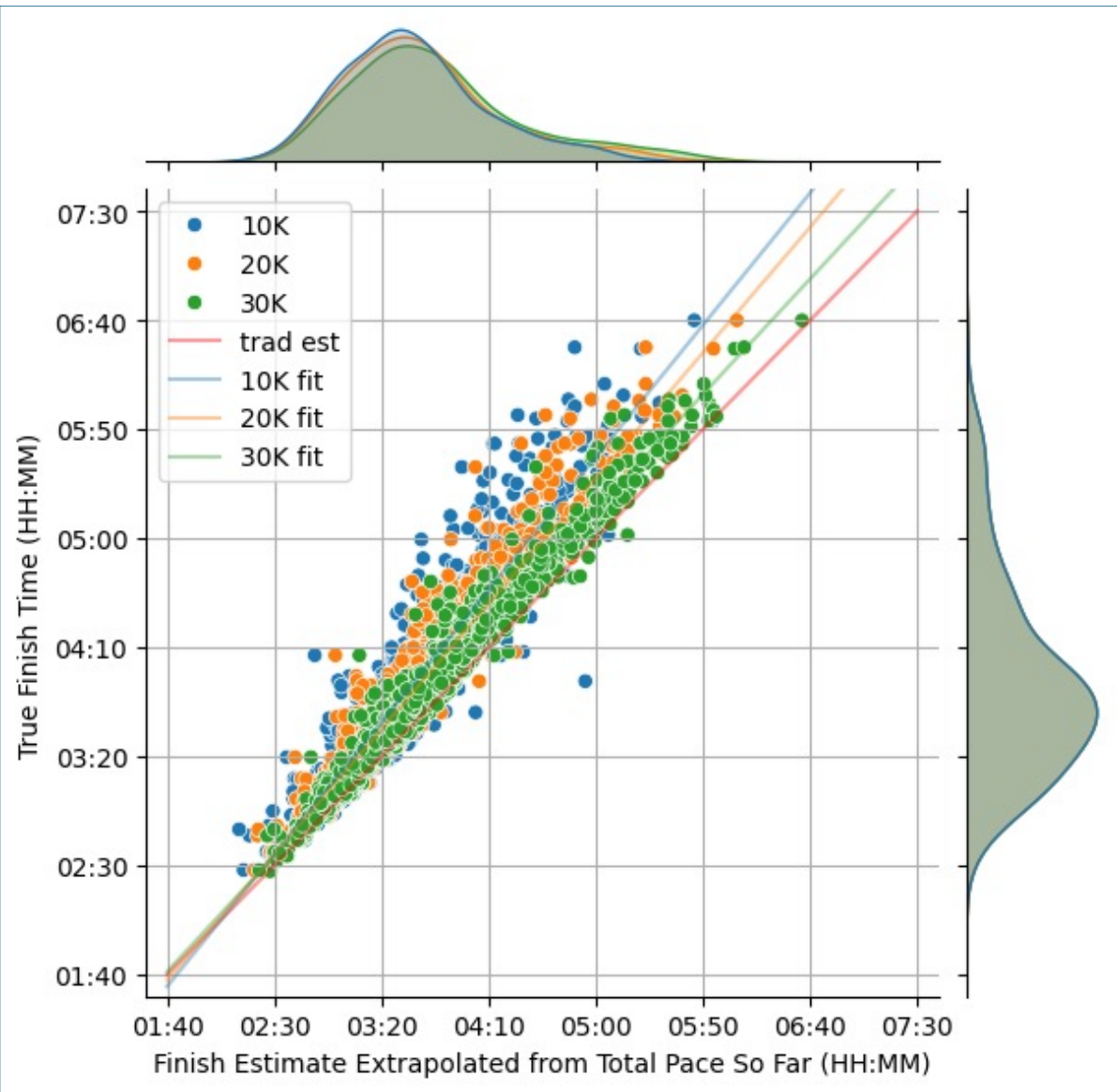
- *Ex: runners usually slow down due to fatigue*

Problem 2: the prediction is a **single point estimate** that has **no additional information about the uncertainty** behind the estimate

- *We should feel more confident in estimate when runner closer to finishing*
- *Predictions can reflect uncertainty*

Problem 1 with Traditional Estimates (visualized)

Problem 1: the estimates don't consider the **in-race context** that can determine if a runner is **likely to finish faster/slower than the observed pace**



- Most points fall above traditional estimate line (red)
- Runners finish slower than current pace so far

Two Approaches to Address The Problems

Bayesian Linear Regression

- Incorporates Bayesian statistics placing priors on parameters of linear regression model
- Posterior of possible finish times
- Use to form credible intervals
- PyMC: Python library to create Bayes models



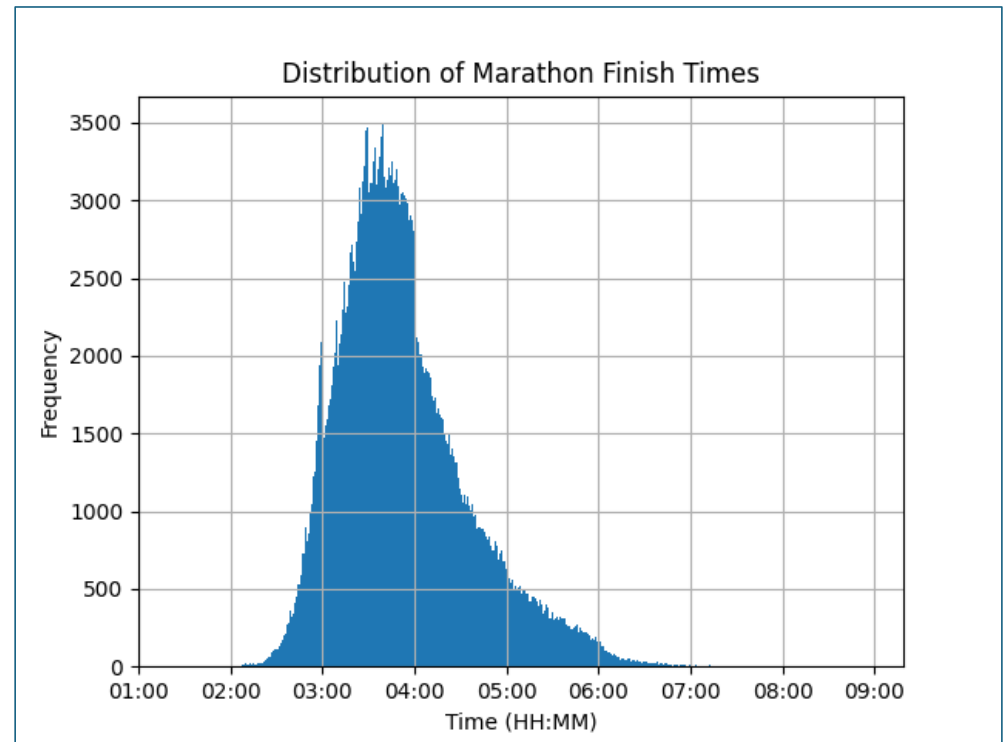
Quantile Regression

- Estimates quantiles of response variable
- Interpret corresponding quantiles as bounds for credible intervals
- Statsmodels Python library



Dataset

- Focus: Boston Marathon
- Every finisher from 2009-2023 (n=312,805)
- Features: name, age, gender, and splits (5K, 10K, 15K, 20K, HALF, 25K, 30K, 35K, 40K, and FINISH)
- Training set: 2009-2022 (n=286,777)
- Test set: 2023 (n=26,028)



Name	Age	M/F	5K	10K	15K	20K	HALF	25K	30K	35K	40K	Finish Net	Year
Merga, Deriba	28	M	874	1769	2685	3620	3819	4570	5459	6382	7284	7722	2009
Rono, Daniel	30	M	874	1770	2686	3622	3820	4570	5460	6411	7348	7772	2009
Hall, Ryan	26	M	874	1769	2685	3621	3820	4570	5483	6418	7366	7780	2009
Kebede, Tekeste	27	M	874	1769	2686	3621	3820	4570	5483	6418	7366	7789	2009
Cheruiyot, Robert	20	M	874	1769	2685	3621	3819	4570	5486	6443	7391	7806	2009
...	...	...	...	...	...	...	...	...	...	...	...	...	...
Davidson, Jonathan	52	M	2166	4327	6689	9054	9562	11466	14322	19043	22652	24056	2023
Arai, Tatsuya	42	M	2104	4184	6508	10485	11118	13388	16546	19452	22649	24188	2023
Foulds, Rhonda	59	F	2462	5102	7854	10700	11388	13699	17116	20494	23582	24994	2023
Weisberg, Scott	53	M	2822	5764	8754	11809	12435	14797	17819	21013	24111	25460	2023
Burros, Aaron	52	M	2503	5339	8183	11247	11888	14287	17575	20993	24424	26019	2023

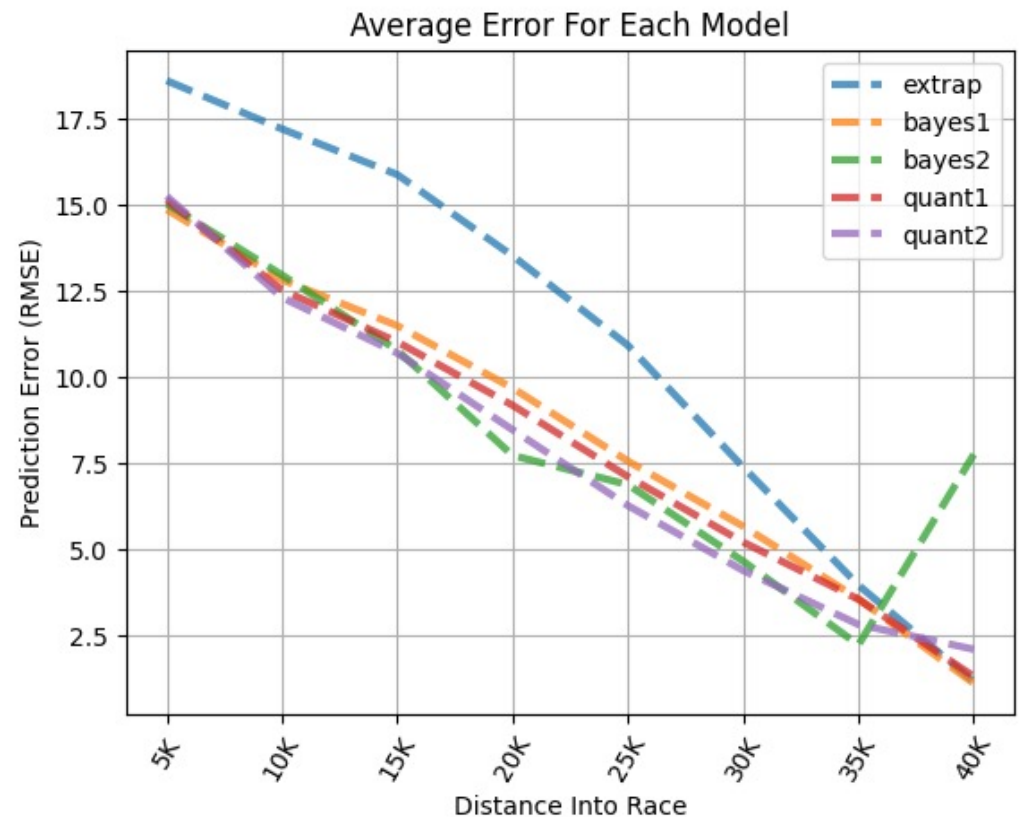
Methods

- **Method A**: the traditional method of extrapolating the current pace
 - **Method B**: Predictor: overall pace so far (*total_pace*)
 - **Method C**: Predictors: overall pace so far (*total_pace*) AND the “current pace” (the pace of the most recent 5K)
-
- ***extrap***: the baseline, traditional extrapolation prediction (**Method A**)
 - ***bayes1***: Bayesian linear regression with **Method B** predictors
 - ***bayes2***: Bayesian linear regression with **Method C** predictors
 - ***quant1***: Quantile linear regression with **Method B** predictors
 - ***quant2***: Quantile linear regression with **Method C** predictors

Results

- Models have very similar RMSE at most levels
- All outperform traditional method, especially at beginning / middle stages of race
- **Method C** models slightly better than **Method B** models overall
- **Method C** models at 40K

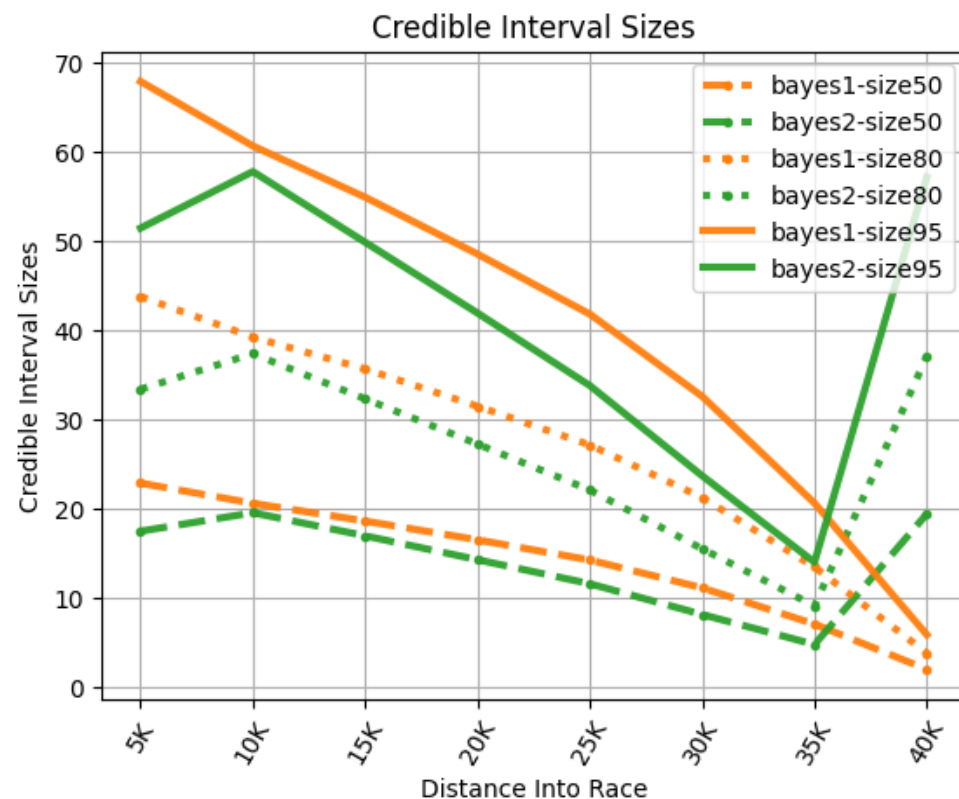
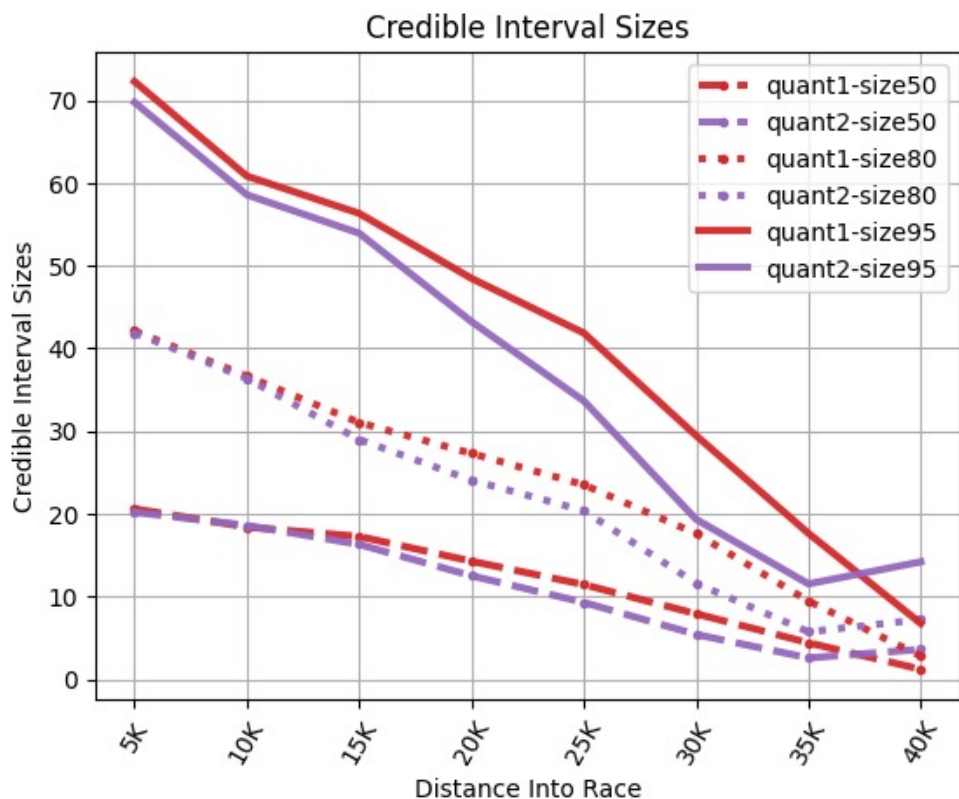
- **bayes1**: Bayesian (**Method B**)
- **bayes2**: Bayesian (**Method C**)
- **quant1**: Quantile (**Method B**)
- **quant2**: Quantile (**Method C**)



Validation – Credible Interval Sizes

We expect credible interval sizes to **decrease** over time

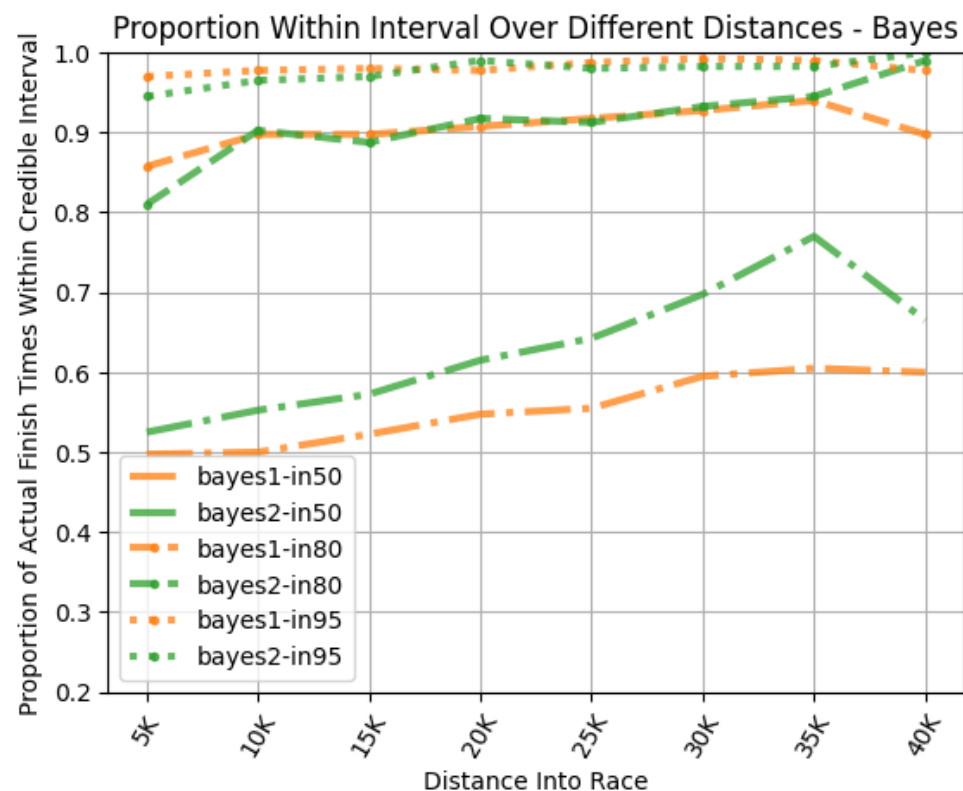
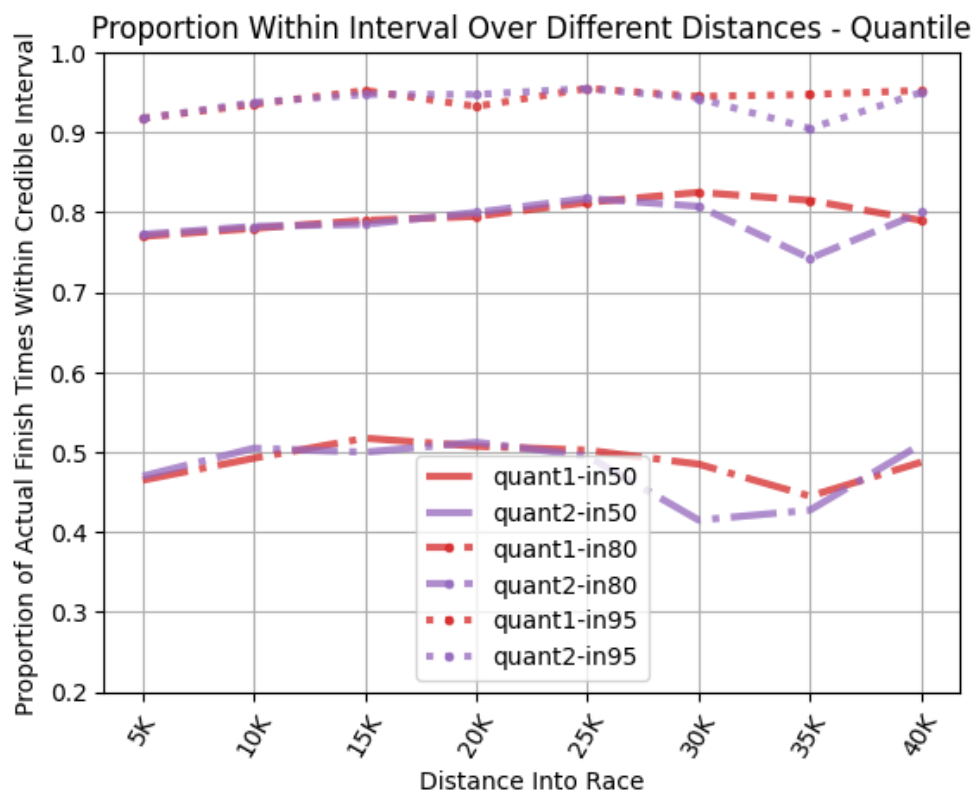
we can create an x% credible interval $[t1, t2]$ such that the true finish time falls between $t1$ and $t2$ x% of the time



Validation – Proportion In Interval

Roughly X% of
runners should finish
in each X% interval

we can create an x% credible
interval $[t1, t2]$ such that the
true finish time falls between $t1$
and $t2$ x% of the time



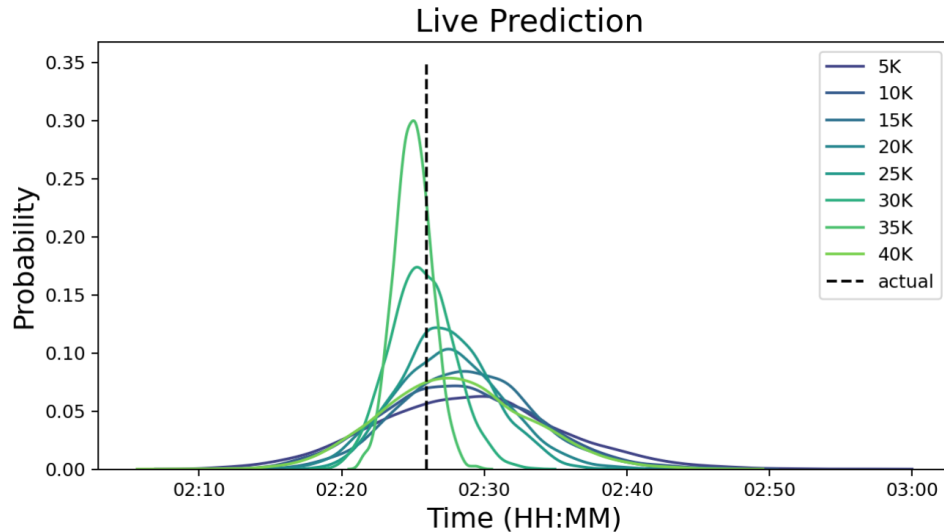
The Application

*Dynamically simulate
races by inputting splits
to get results in real time*

[https://bonyejekwe.
shinyapps.io/
marathon_predictor](https://bonyejekwe.shinyapps.io/marathon_predictor)



Live Prediction: Plot



Live Prediction: Table

Actual finish time: 02:25

dist	median	range_50	range_80	range_95
5K	02:28	02:24-02:32	02:20-02:37	02:16-02:42
10K	02:28	02:24-02:31	02:21-02:35	02:18-02:39
15K	02:28	02:25-02:31	02:23-02:34	02:20-02:38
20K	02:27	02:24-02:30	02:22-02:32	02:20-02:35
25K	02:27	02:25-02:29	02:23-02:31	02:21-02:33
30K	02:25	02:24-02:27	02:22-02:28	02:21-02:30
35K	02:24	02:24-02:25	02:23-02:26	02:22-02:27
40K	02:27	02:24-02:31	02:21-02:34	02:18-02:38

Conclusions, Limitations and Future Work



- Models outperform traditional method
- Choosing better predictors (age, gender, experience)
- Implementing more models (only bayes1 in app)
- Predicting for more races (other than Boston Marathon)
- Adapt model towards different goals (ie: predict at 30K mark)